The Chinese Experience of Eliminating Hunger

The world food problem is one of the most complex problem in the world, which worries many people across the world(1). In 1996,186 countries vowed to reduce global hungry population to 400 million cooperatively, yet so far, the complexity of the problem has made the goal still far from being reached(2).

In China, the huge population makes food supply essentially important. In history, Chinese people, especially who were at the bottom of the society, suffered great much from hunger. For example, the Great China Famine during 1959 to 1961 killed 27.91 million across the mainland China. However, since the Chinese economic reforms which started in 1978, everything has changed. In 1990, 23.9% of the mainland Chinese were starving, while in 2016, only 9.3% of them suffered from hunger (2). Considering the enormous population of China which takes up about one fifth of the world population, this progress is remarkable. How China is eliminating hunger is an important part of the wider global actions. Learning from the Chinese experience, which consists of two major parts including increasing productivity and marketisation, can be very helpful for fighting hunger across the world.

Increasing productivity

To increase agricultural productivity, China puts efforts into 4 major aspects.

The first aspect is disaster reduction and prevention. In the past, the weather is a pre-dominant factor which leads to agricultural failures from time to time; therefore, the Chinese government has taken measures to change this. Modern agricultural infrastructures have been built across the country, in which reservoirs and dams can take up a large proportion.

The second aspect is the farmland management. In the 1970s, the Chinese government started to focus on so-called farmland capital construction, which included land clearance, terraced field construction, soil quality improvement (including soil pH adjustment), farmland shelterbelt network building, irrigation optimiza-tion, water conservancy and agriculture mech-anization (3), which has helped to improve the resistance to disasters, increase the farmland area and boost the land productivity, contrib-uting to the increase of the total agricultural production.

The third aspect is the agricultural technologies. First of all, new plant strains have been developed for planting. Among them, the super hybrid rice which was first developed by Yuan Longping has boosted the productivity of rice to about 12,000kg/hm^2(4). The introduction of new fertilizers and pesticides has improved the yield largely. New agricultural techniques like multiple cropping also helped a lot (5).

Lastly, the agricultural policies play a huge part. China introduced crop purchase and storage system in 1977, which allows the government to buy crops from farmers and sell it when the prices are going unaffordable. As a matter of fact, this system also ensures that farmers can make profits by farming, as the farmers can sell the crops at a price no less than the governmental prices.

In the early 1980s, China allowed the household-responsibility system, which emerged in 1977, to replace the Soviet-styled collective ownership which allocated all products evenly regardless of differences in the individual contribution; thus, farmers began to take a more active part in farming(6).

Since 1978, the productivity of farming contin-ue to grow rapidly until now, with the total production of grain doubled (2). As the eco-nomics become more and more market-driven, specialized farming focusing on local advantages, along with the free trade and fast transport, make Chinese consumers have easier access to different kinds of food from all parts of the country at affordable prices.

Opening-up policy

Apart from the efforts to increase productivity, China's opening-up policy, which allows more trade with foreign countries, also plays an important role in agriculture in China. The Chinese market of the import and export of agricultural products has been of important great importance to the world. Importing food is vital to China, for it is predicted by 2020, China's grain import will have increased from 70 million tons in 2012 to over 100 million tons to satisfy the growing domestic need. (7)

Chinese agricultural trade

The Chinese government set up a series of policies to fully open the agricultural market, which includes two major goals.

First, China gradually reduces the agricultural tariffs and phases out non-tariff measures which limits the import of grain, vegetable oil, cotton, sugar, wool and other key agricultural products. Nowadays, China has become one of the countries which own smallest amount of tariffs on the products.

Second, the government works on the implementation of the strategy of "going global" in agriculture to support Chinese companies invest overseas and increased the intensity of foreign cooperation in agriculture. Several agricultural enterprises, including private enterprises, actively invest in Southeast Asia, Africa, South America and other regions to operate agricultural projects(8).

Positive impact of the opening-up

Ever since 1978, China has accomplished most of her agricultural policy objectives. The annual growth of agricultural GDP was around 7% during 1979-1984. Moreover, despite the declining number of farmers and the decreasing area of farmland and 1997 Asian financial crisis, the rate remains near 4% (9). From 1980 to 1995, the value of agricultural exports increased by 7.1%

averagely and the agricultural import value grew at an average rate of 5.9 %(10). Since 1978, the import of foreign food continues to grow, making China among the largest and the most influential markets (7).

Lessons from the opening-up policy

The development of agriculture is essential to hunger reduction, but without a open market, the food will hardly get to where it is most needed easily. For a developing country, an open food market needs an active but careful government to sustain the market. The market will help eliminate hunger if it runs well. However, at least in China, this is not the primary solution, for China can self-sustain the rice for daily meal on her own and the imported food takes a less dominant position(7). improving agricultural productivity is still urgent for those suffering from hunger.

Chinese agriculture in Africa

China has done a great job in eliminating hunger within her land by producing and trading more agricultural products. In the meantime, she is also trying to export Chinese experience in agriculture to other countries, especially to African countries. Since 1950s, China has begun to expand development projects in Africa which included efforts to improve agri-culture in Africa, introducing Chinese solutions to African problems. Not until the 1970s, China had had more foreign aid projects in Africa than the United States (10).

There are two major phases of Chinese aids to African agriculture, each of which had different influences: the politics-driven aid before 1978; the market-driven aid since Deng decided to open up China to the world in 1978 (11).

Irrigation infrastructure construction and management were transferred to Africa in the early Chinese aid projects, along with the intensive input method of agriculture including mechanization and cultivation practices, which was hardly known to Africans (11). However, the input, including the infrastructures like irritation systems and practice like using chemical and organic fertilizers, was often unsustainable when Chinese experts left (12). But in certain countries like Senegal, high-yielding semi-dwarf varieties of rice that the Chinese from either the mainland or Taiwan introduced has become part of local farming since (13). Nevertheless, although the output of agriculture in Africa climbed stably, the food was still not enough to feed the booming population (14).

Since China began economic reforms in 1978, the Chinese has made the aid more market-oriented, with technocratic officials to take a dominant position (12). Funds, which are most needed for agricultural modernization in Africa, along with Chinese companies, are coming to African agriculture, especially when the aid that used to be mainly from the US and the Soviet has dropped since the cold war has ended (15). Enterprises began to appear in the Chinese Agricultural Technology Demonstration Centers in Africa, where Chinese experts trained local farmers (16) and officials (17).

However, although commented as “advanced technologies” by many African officials and associated with the enthusiasm to reach self-reliance of food in the countries by the media, the Chinese hybrid rice and agricultural machine never find an easy way into the markets, due to the resistance of the local farmers (12). Yet there are successful examples of such aids, such as Benin–China cooperation in the development of Ouémé valley, which has built an irrigation system in the newly developed area, believed to be helpful for Africa to explore agricultural potential (18).

The population affected by hunger in Sub-Saharan African countries decreased from one third in 1991 to less than one fifth in 2015 (19). During the time, China surely made a contribution, yet to eliminate hunger in Africa, there is more to be done in the future.

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